

M000 · Changing a Variable — English Worksheet

Topic: Changing a Variable While You Build · **Course:** Builder Basics · **Time:** about 45 minutes

1 · Predict 🧠

A variable can **change while the code runs**. `height = height + 1` means: take what `height` holds, add 1, put it back.

Read each set of steps. Write what you think will happen.

```
height = 1
repeat 4 times:
  place block down
  move up by height
  move forward
  height = height + 1
```

Does each step climb the same amount, or more each time? Say why.

Fill the table. Start at 1 and add 1 after every step.

Step	height before the step
1	1
2	
3	
4	

```
gap = 7
repeat 4 times:
  place block down
  move forward by gap
  gap = gap - 2
```

Write the four values of **gap** . Do the blocks get further apart or closer together?

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, rewrite it, then explain why the original was wrong.

Bug A — Each step should climb higher than the one before.

```
height = 1
repeat 4 times:
    place block down
    move up by height
    move forward
```

Write the fixed code:

What shape do you get instead? Why does your fix work?

Bug B — The first step should climb 1, then 2, then 3, then 4.

```
height = 1
repeat 4 times:
    height = 1
    place block down
    move up by height
    move forward
    height = height + 1
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The first step should climb **1**, not 2.

```
height = 1
repeat 4 times:
    height = height + 1
    place block down
    move up by height
    move forward
```

Write the fixed code:

Why does the order of those two lines matter?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a **staircase** where each step climbs higher than the one before, using a variable that changes inside the loop. Then change **one number** — the start value or the amount you add — and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does. Point at the line that changes your variable.

2 · Your build. Point the camera. Show both staircases.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.